

Entity Resolution with FAISS + Splink

One system, three conversations

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How to use this deck

Part	Audience	Focus	After the talk
1	C-suite	Value, risk, cost	Decision / review
2	Engineering leadership	Architecture, design, roadmap	Carry it forward
3	Line engineers	Orientation, stack, hands-on	Go hands-on

Line engineers: this orients you. The whitepaper ([docs/entity_resolution_whitepaper.pdf](#)) is the source of record; the numbers here come from it.

Part 1 · Executive Summary

The 60-second version

The problem

Organisations hold **many records for the same real person**, and they disagree.

- Names differ (e.g., `R. Martinez` vs `Robert Martinez`).
- Addresses change; emails and dates go missing (**30% missing** here).
- Duplicates fragment CRM, risk, and reporting.

Wrong answers are expensive: linking two *different* people is usually far worse than missing two records of the same person.

The solution: two stages

Fast retrieval + explainable matching.

```
Query → [embed] → [FAISS top-k]
      → [trained scorer] → Match / No match
```

- **Blocking** (FAISS): cheaply finds the few most likely matches.
- **Linkage** (trained scorer): weighs the evidence per field and returns a probability, not just yes/no. Splink trains the `m/u` ; a lightweight scorer serves them per query.

What it delivers

- **99.3% F1** on the synthetic test corpus (15,000 labelled queries).
- **~18 ms/query amortized *batched* throughput**; the honest **online** per-query path is **~38 ms median** (embedding + FAISS + a lightweight trained scorer — no per-query Splink construction).
- **Persisted and reusable**: the index reloads without re-embedding the population.
- **Defensible numbers**: every result reproduces by re-running the code.

Set the risk dial, not a magic number

Splink returns a **probability**; a pair is called a match when that probability is at or above a threshold τ . One number encodes the business trade-off.

- **Raise** τ → fewer false positives (fewer *wrong* links) at slightly more false negatives.
- **Lower** τ → catch more true matches, at more false positives.

The cost-based rule:

$$\tau^* = C_{FP} / (C_{FP} + C_{FN})$$

If a **false link** (FP) is 20× more costly than a **missed link** (FN), then

$\tau^* = 20/21 \approx 0.95$ — the model is tuned to the cost of each mistake. (Valid under calibrated probabilities; with labels, sweep τ on validation — whitepaper Appendix — Deriving τ .)

One parameter lets a risk-averse line (fraud/AML) and a recall-first line (marketing) share the same model, each tuned to its own economics.

Cost and scale

- The 50,000-record index is **~87 MB** today.
- Capacity is predictable from RAM;

RAM	Recommended capacity
16 GB	~6.9 M records
32 GB	~13.8 M records
128 GB	~55.1 M records

Around **7–14 M records** (or when you need replicated, multi-region durability) buy a **vector database**; before that the in-process index is the lean choice.

The honest caveats

- Results are measured on **synthetic data**; production needs a **labelled sample of true duplicates** to calibrate and validate.
- **Calibrate `m/u` with labels before production** — supervised calibration improves precision (→100% here); do it with a **pinned prior** so the EM route's free prior can't shift scores (the whitepaper's calibration-pitfall).
- Volume must clear our capacity thresholds before any external-infra spend.

The ask

- Sponsor a short **proof-on-real-data** phase: a representative, labelled sample of the population we must link (with access/privacy).
- We then commit to an F1 on *that* data, not the synthetic one.

Value in one line: near-perfect, explainable identity resolution with batched throughput ~18 ms/query and **online per-query latency ~38 ms** — the per-query Splink rebuild is gone, replaced by a lightweight trained scorer (whitepaper §3.3).

Part 2 · Engineering Leadership

Architecture, measurements, where to invest

Architecture at a glance

Query $\rightarrow$ MiniLM embed $\rightarrow$ FAISS top-k $\rightarrow$ lightweight scorer $\rightarrow$ $p(M \mid \gamma)$

▲
name / DOB / email / address comparisons, τ threshold

- Embeddings are **L2-normalized**, so FAISS inner product = cosine.
- **Fellegi–Sunter**: per-field weight $w = \log_2(m/u)$, combined with a match prior, compared to τ .
- **Train with Splink, infer with a lightweight scorer**: Splink trains the m/u once; each query is scored by a small weight-table scorer (Splink's comparison SQL, precompiled) — no per-query Splink **Linker** or DuckDB pipeline. The scorer matches Splink's posteriors to $\sim 1e-7$.
- A store abstraction lets us swap the in-memory index for an **external vector DB** later without changing blocking or linkage.

Where performance comes from

Blocking recall is the ceiling; the scorer is where F1 is won or lost.

- On the confusion-matrix query set (15k queries): top-20 blocking recall **98.87%** → end-to-end recall **98.83%** — the linkage stage rejects only 4 of the 9,887 retrieved positives; blocking is the binding constraint.
- A "perfect" blocker would take F1 from **99.30** → **~99.87%**, since 113 positives were missed at top-20; retrieval now has real headroom.
- (Section 7, a different query construction: blocking recall 99.75%; compact raises it to 99.88%.)

Design implication: improving the embedding/blocking engine is **not** the lever today; linkage configuration is.

The measured levers

Lever	Effect	Verdict
Threshold τ	0.85→0.95: F1 barely moves (99.30→99.34%); recall scarce	Small—retrieval is the lever now
Serialization	single seed: recall 99.88%, F1 99.83%; five-seed mean F1 99.90 vs 99.82 (+0.08)	Yes—small, reproducible
Blocking size k	20→100: +recall, $\sim 2.4\times$ scorer time	Diminishing returns
Weaken address	never beat full-strength	No (this data)
Retrain m/u	§8.1: supervised improves (98.44 vs 97.64, precision→100%); §8.2 100k: EM at default prior best (97.85%)	Yes for supervised; EM needs prior pinned

τ is the cost dial (engineering view)

Splink's decision rule: classify as *match* when $P(M|\gamma) \geq \tau$ (range $\sim[0.5, 0.999]$).

- **Raise τ** → fewer false positives, at more false negatives.
- **Lower τ** → more true matches, at more false positives.

Decision-theoretic setting: $\tau^* = C_{FP} / (C_{FP} + C_{FN})$ — presumes a calibrated posterior, zero cost for correct decisions, and per-pair decisions (whitepaper Appendix — Deriving τ).

- Measured: τ 0.85 → 0.95 moved F1 barely (99.30% → 99.34%; FP 22→4) because recall — not threshold — is the scarce resource.
- With recall headroom, raising τ is the cheapest precision lever — the mechanism to encode FP/FN business costs. (Methods: whitepaper **Appendix — Deriving τ** .)

Calibration: stable and worth doing

Retraining the match model is **worth doing with real labels** — and stays coherent when the prior is handled properly.

1. **Supervised calibration improves results** (§8.1): fitting `m/u` on labelled match/non-match pairs lifts F1 to **98.44% vs 97.64% untrained**, precision to **100%** (FP 62→0) at unchanged recall — under the lightweight scorer's weight tables.
2. **EM matches or beats untrained** in every measured configuration (including the duplicate-bearing set where all variants reach F1 1.0 at $\tau=0.85$): calibrated `m/u` never underperform the defaults at a fixed operating point.

Supervised calibration uses genuine match/non-match evidence; EM fits `m/u` from the (resemblance-biased) candidate structure. Both stay stable when the prior is kept coherent.

Rule: the default config is a coherent bundle (weights + prior + τ). Change any part and re-validate **all three** on held-out data.

Keep the prior anchored (default or blocking-adjusted estimate), fit `m/u` with the prior pinned, then **tune τ jointly**, with a coherence guard. Full algorithm: **Appendix — Tuning the Match Prior**.

Capacity and when to buy infra

- **Memory is the binding constraint** (~1,746 bytes/record).
- Move to an **external vector DB** around **7–14 M records**, or when you need shared/multi-region **durability and replication**, or beyond **~50 M** as a rule of thumb.
- The persisted index reloads **without re-embedding**, so redeploy and multi-consumer serving are cheap.

Operational readiness

- Persisted, reload-friendly store with `update` / `delete` ; external-store interface is the vector-DB swap point.
- **Reproducible**: every experiment maps to a script; scripts accept `--input` .

Roadmap

1. Invest in **linkage calibration** (labelled pairs + joint τ /prior tuning).
2. Do **not** spend on blocking-engine research at this scale. (Future: canopy/embedding-blocked `m/u` training — §Further Research — but only where blocking recall is the binding constraint.)
3. Add a **CI gate** locking F1 on a held-out labelled set.
4. **Gate go/no-go on internal corporate customer data** — the NC-voter run is only a public-data sanity check (pre-deduplicated; no DOB/email), so the production decision uses the org's own labelled data.

Part 3 · Line Engineers

Orientation — where the knobs are

What the code does

```
Blocker    : embed query → FAISS top-k candidates (cosine)
Scorer     : Splink-trained weight tables → per-candidate match probability
            (Splink comparison SQL, precompiled; no per-query Splink Linker)
Store      : add / update / delete / save / load (no re-embed on load)
```

- Match probability = Bayesian combination of per-field weights $w = \log_2(m/u)$, a match prior, thresholded at τ .
- Splink is the **training** engine (estimates m/u and the prior); the lightweight scorer serves them at query time and matches Splink's posteriors to $\sim 1e-7$.
- In-memory index today; `IndexingStrategy` / `VectorDatabase` is the seam for a vector DB later.

Scorecard to remember

- Confusion matrix (5k refs / 15k queries): **F1 99.30%**, recall 98.83%, precision 99.78%.
- Same-set blocking recall: **98.87%** @k=20 (Section 7's query set: 99.75%; compact 99.88%).
- **~18 ms/query amortized batched** (embed + block + scorer); **cold per-query online path median ~38 ms** on the 50k index (embedding ~24 ms + FAISS ~25 ms + scorer ~11 ms).
- NC-voter (real schema, synthetic mutations): **F1 92–94%**; blocking is the binding constraint at the default **k=20** (linkage becomes binding at **k=100**).

Baseline engineering numbers — config, hardware, and data shape all move them.

Reproducing experiments

Run from repo root; scripts have `--help` and deterministic seeds (default 42).

Paper section	Script
§7 evaluation	<code>scripts/experiment_section7_eval.py</code>
§8 confusion matrix	<code>scripts/experiment_confusion_matrix.py</code>
§8.1 m/u calibration	<code>scripts/experiment_mu_calibration.py</code>
§8.2 duplicate benchmark	<code>scripts/experiment_duplicate_benchmark.py</code>
§8.3 F1 sweep	<code>scripts/experiment_f1_sweep.py</code>
calibration robustness	<code>scripts/experiment_mu_tau_interaction.py</code>
joint $\tau \times$ prior surface	<code>scripts/experiment_mu_prior_tau_surface.py</code>
NC-voter replication	<code>scripts/ncvoter/*</code> + <code>extract_ncvoter.py</code>

For **your** data: population-based scripts accept `--input-records FILE`.

Gotchas — tuning & thresholds

- **Calibrate m/u with labels — and pin the prior.** Supervised calibration improves F1 (98.44 vs 97.64) and precision ($\rightarrow 100\%$); the pitfall is the EM route's free prior, which underperforms unless anchored. Re-validate prior + τ + weights together (Appendix — Tuning the Match Prior).
- **τ encodes business cost.** $\tau^* = C_{FP} / (C_{FP} + C_{FN})$; exposed via `--threshold` / `--tau` and the F1 sweep. Raise τ for fewer wrong links, lower it for recall-first (Appendix — Deriving τ).
- **Tune the prior properly.** Anchor λ , fit with the prior pinned, tune τ jointly, watch the score-shift guard (Appendix — Tuning the Match Prior).

Gotchas — blocking & data

- **Blocking recall is a hard ceiling.** If recall is low, raise `k` or serialization, not the embedding model.
- **Address weakening didn't help** here — test, don't assume.
- **Real data needs a mutation/duplicate model** to score (`scripts/ncvoter/ncvoter_util.py`).
- **NC voter has no full DOB and no email** — those comparisons are weak there.

Where the details live

Whitepaper: `docs/entity_resolution_whitepaper.pdf` (source: `.tex`).

- **§3 Two-Stage** — blocking/linkage contract and costs; **Address volatility & temporal decay** (§3.5) consolidates why address evidence changes, the continuous retrieval-time decay and the bucketed comparison-level alternative, and their measured impact.
- **§7 Evaluation** — recall, thresholds, latency, ablations.
- **§8, 8.1, 8.2, 8.3** — headline results and the "don't retune alone" evidence.
- **§9 NC-voter** — real schema, mutation model, `k` scaling.
- **Appendix: Deriving τ** — cost / F1 / GMM / transitivity methods.
- **Appendix: Tuning the Match Prior** — anchor `$\lambda$` , then tune `( $\lambda$ ,  $\tau$ )` jointly (keep the prior coherent across retraining).
- **Use of AI** — disclosure and verification.

Checklist before you leave

1. Run `scripts/experiment_confusion_matrix.py --count 5000` and read the JSON.
 2. Reproduce the stable calibration result: `scripts/experiment_mu_tau_interaction.py --base-count 5000`.
 3. Swap `--input-records <your file>` into a population-based experiment.
 4. Find where **blocking recall** caps your F1 — then plan linkage work.
- That is the fastest way to make these results your own.

Thank you

FAISS + Splink — one pipeline, three audiences

Bring a labelled sample of your real duplicates.